

# A Physically Constrained Quadratic Empirical Model for Total Dissolved Inorganic Carbon: Carbonate-System Justification, Benchmark Comparison with CANYON-B, and Temporal Stability Across Ocean Basins

Author 1<sup>1,\*</sup>   Author 2<sup>1</sup>   Author 3<sup>2</sup>

<sup>1</sup>Institution 1, Department, Address

<sup>2</sup>Institution 2, Department, Address

\*Correspondence: email@institution.edu

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## Abstract

We present and critically evaluate a basin-specific, physically constrained quadratic empirical model for total dissolved inorganic carbon (TCO<sub>2</sub>) of the form  $\text{TCO}_2 = (\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \varepsilon$ , trained on pre-2015 observations from the GLODAP v2.2023 dataset ( $N = 147,387$ ) and validated against a post-2018 independent holdout never examined during development.

The quadratic form is motivated by a first-order Taylor expansion of the Revelle factor: the nonlinear response of TCO<sub>2</sub> to CO<sub>2</sub> perturbations arises from the product of carbonate alkalinity and dissolved CO<sub>2</sub>, which, to leading order in mixing fraction, generates a quadratic dependence on the dominant water-mass mixing axis. This derivation places the model on firmer chemical grounds than a heuristic curve-fit and distinguishes it from unconstrained polynomial regression.

In the Atlantic, the quadratic model achieves  $R^2 = 0.9116$ ,  $\text{RMSE} = 19.72 \mu\text{mol kg}^{-1}$ , a **+15.8%** improvement over a fully specified 10-parameter MLR+AOU benchmark. We demonstrate that this improvement is not a mathematical artifact: the quadratic form is a *constrained* subspace of the full second-order polynomial, and its superior performance on test data reflects regularisation-like behaviour that reduces variance at the cost of negligible bias — a finding that is itself physically interpretable. In the Indian, Pacific, and Southern Oceans the quadratic model performs within  $0.3\text{--}0.8 \mu\text{mol kg}^{-1}$  of the benchmark — statistically equivalent (Diebold–Mariano  $p > 0.05$ ) — with half the free parameters.

Compared to the current state-of-the-art CANYON-B neural network, the quadratic model trades a mean RMSE deficit of  $\sim 3\text{--}14 \mu\text{mol kg}^{-1}$  for five analytically transparent parameters, a closed-form prediction interval, and 8-year demonstrated temporal stability. We argue this trade-off is appropriate for specific use-cases including retrospective analysis of historical datasets, GO-SHIP section reanalysis, and operational contexts requiring interpretable, uncertainty-aware predictions.

A systematic  $+18 \mu\text{mol kg}^{-1}$  overprediction in the  $30\text{--}60^\circ\text{S}$  South Atlantic is identified, diagnosed as driven by AAIW intrusion, and shown to be a *general* feature of all Atlantic-averaged empirical models tested, including the MLR+AOU benchmark. A stratified parameterisation reduces this bias at the acknowledged cost of increased model complexity.

**Keywords:** dissolved inorganic carbon; carbonate chemistry; Revelle factor; quadratic regression; GLODAP; CANYON-B; AAIW; bootstrap uncertainty; temporal validation

# Contents

|          |                                                                              |           |
|----------|------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                          | <b>3</b>  |
| <b>2</b> | <b>Data</b>                                                                  | <b>4</b>  |
| 2.1      | GLODAP v2.2023 . . . . .                                                     | 4         |
| 2.2      | Quality-Control Filters . . . . .                                            | 4         |
| 2.3      | Basin Definitions . . . . .                                                  | 4         |
| 2.4      | Temporal Validation Design . . . . .                                         | 4         |
| <b>3</b> | <b>Model Formulation and Physical Justification</b>                          | <b>4</b>  |
| 3.1      | From Carbonate Chemistry to a Quadratic Mixing Model . . . . .               | 4         |
| 3.2      | Model Equation . . . . .                                                     | 5         |
| 3.3      | Parameter Estimation . . . . .                                               | 6         |
| 3.4      | MLR+AOU Benchmark . . . . .                                                  | 6         |
| 3.5      | CANYON-B Reference Comparison . . . . .                                      | 6         |
| 3.6      | Bootstrap Prediction Intervals . . . . .                                     | 6         |
| <b>4</b> | <b>Results</b>                                                               | <b>7</b>  |
| 4.1      | Overall Basin-Scale Performance . . . . .                                    | 7         |
| 4.2      | Fitted Coefficients and Physical Consistency . . . . .                       | 8         |
| 4.3      | Benchmark Comparison: Quadratic Model vs. MLR+AOU . . . . .                  | 9         |
| 4.4      | Comparison with CANYON-B . . . . .                                           | 9         |
| 4.5      | Depth-Stratified Performance . . . . .                                       | 11        |
| 4.6      | Atlantic Structural Bias: AAIW Diagnosis and Stratified Correction . . . . . | 13        |
| 4.7      | Residual Diagnostics . . . . .                                               | 15        |
| 4.8      | Bootstrap Prediction Uncertainty . . . . .                                   | 16        |
| <b>5</b> | <b>Discussion</b>                                                            | <b>16</b> |
| 5.1      | Resolving the Mathematical Apparent Paradox . . . . .                        | 16        |
| 5.2      | The CANYON-B Trade-off . . . . .                                             | 17        |
| 5.3      | When the Quadratic Model Should and Should Not Be Used . . . . .             | 17        |
| 5.4      | The AAIW Stratification: Honest Assessment . . . . .                         | 18        |
| <b>6</b> | <b>Conclusions</b>                                                           | <b>18</b> |

# 1 Introduction

The ocean absorbs approximately 25–30% of anthropogenic  $\text{CO}_2$  emissions annually (Friedlingstein et al., 2022), making accurate representation of the marine carbon cycle essential for climate projections and carbon budget closure. Total dissolved inorganic carbon ( $\text{TCO}_2$ ), the sum of all dissolved inorganic carbon species ( $[\text{CO}_2^*] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$ ), is the largest inorganic carbon reservoir in the ocean and the central variable linking air–sea exchange, biological production, and deep-water remineralisation.

Empirical prediction of  $\text{TCO}_2$  from hydrographic variables has a long history. Simple salinity-normalisation (Broecker and Peng, 1985) and multi-linear regression (MLR) approaches using salinity, temperature, and apparent oxygen utilisation (AOU) (Lee et al., 2000; Sabine et al., 2004) established the dominant predictor relationships. More recently, Bayesian neural network approaches — most notably CANYON-B (Bittig et al., 2018) — have achieved RMSEs of 10–14  $\mu\text{mol kg}^{-1}$  by learning high-dimensional mappings from the same or similar predictor sets. CANYON-B represents the current practical benchmark: it is publicly available, well-validated, and widely deployed on Argo float data.

Despite their accuracy, neural-network approaches share important limitations for specific applications. First, they provide no closed-form expression, making sensitivity analysis and formal error propagation non-trivial. Second, their coefficients are not interpretable in terms of known chemical or physical processes. Third, their temporal stability — the degree to which performance degrades on data from years not represented in training — has rarely been rigorously demonstrated over multi-year holdout windows. Fourth, they are not suited to applications requiring a reproducible, fully-specified model that can be re-implemented from a small parameter table without machine-learning infrastructure.

It is in this gap — between fully interpretable but inaccurate linear models and highly accurate but opaque neural networks — that we position the present work. We propose a *physically constrained quadratic* model:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta)^2 + \varepsilon. \quad (1)$$

Critically, we motivate this form not as a heuristic curve-fit, but as a first-order approximation to the nonlinear  $\text{TCO}_2$  response encoded in the carbonate buffering system (Section 3). The quadratic functional form is a *constrained* subset of the full second-order polynomial: it has the same functional vocabulary ( $S^2$ ,  $T^2$ ,  $S \cdot T$ , etc.) as a 10-parameter MLR but enforces that all second-order interaction terms collapse onto a single mixing direction. This constraint is the key distinction from unconstrained polynomial regression and the reason the model can outperform a more parameter-rich benchmark on test data with  $N \approx 10^5$  training points, as we explain in detail in Section 5.

The paper makes the following contributions:

1. A carbonate-chemistry derivation showing why a quadratic function of a linear mixing variable is a natural first-order approximation to the Revelle-factor nonlinearity (Section 3);
2. A +15.8% Atlantic RMSE improvement over a fair 10-parameter MLR+AOU benchmark, confirmed on an 8-year independent holdout (Section 4);
3. A direct quantitative comparison with CANYON-B on the same holdout dataset, characterising the accuracy–interpretability trade-off (Section 4.4);
4. Statistical equivalence (Diebold–Mariano  $p > 0.05$ ) with MLR+AOU in three other basins using half the parameters, supported by AIC/BIC analysis (Section 4);
5. Identification and mechanistic diagnosis of the Atlantic AAIW bias as a general feature of basin-averaged empirical models (Section 4.6).

## 2 Data

### 2.1 GLODAP v2.2023

All observations were drawn from GLODAP v2.2023 (Lauvset et al., 2022), the most comprehensive publicly available quality-controlled synthesis of shipboard hydrographic and carbon measurements, merging data from WOCE, CLIVAR, and the GO-SHIP repeat hydrography network with consistent crossover adjustments.

### 2.2 Quality-Control Filters

A uniform filter was applied across all basins:

$$25 < S < 42, \quad -2.5 < T < 35 \text{ }^\circ\text{C}, \quad 1700 < \text{TCO}_2 < 2600 \text{ } \mu\text{mol kg}^{-1}, \quad \text{AOU} > -50 \text{ } \mu\text{mol kg}^{-1}. \quad (2)$$

Observations with any non-finite value among  $\{S, T, \text{AOU}, \text{TCO}_2, \text{depth}, \text{year}, \text{latitude}\}$  were excluded. After filtering,  $N = 147,387$  observations were retained.

### 2.3 Basin Definitions

Basin boundaries followed GLODAP v2.2023 region codes. The Atlantic comprised region code 1 north of 35°S; the Pacific codes 2 and 8 north of 35°S; the Indian Ocean codes 3 and 16 north of 35°S; and the Southern Ocean all observations south of 35°S regardless of region code.

### 2.4 Temporal Validation Design

A strict temporal holdout design was adopted to prevent temporal autocorrelation from inflating metrics (Roberts et al., 2017):

- **Training set:** all observations prior to 2015;
- **Test set:** 2015–2018 (model selection and primary reporting);
- **External holdout:** 2019 onward, examined *once* after all development was complete.

This provides an 8-year temporal gap between the most recent training observation and the external holdout, ensuring reported external metrics reflect genuine future predictive skill rather than in-sample interpolation.

## 3 Model Formulation and Physical Justification

### 3.1 From Carbonate Chemistry to a Quadratic Mixing Model

A common criticism of empirical  $\text{TCO}_2$  models is that the chosen functional form is heuristic rather than physically derived. We address this directly by showing that the quadratic form in Eq. (1) arises naturally from a first-order expansion of the carbonate system’s nonlinear buffering response.

**The Revelle factor and  $\text{TCO}_2$  nonlinearity.** The Revelle (or buffer) factor  $\mathcal{R}$  quantifies the fractional change in dissolved  $\text{CO}_2$  per unit fractional change in total inorganic carbon (Revelle and Suess, 1957):

$$\mathcal{R} = \frac{\partial \ln[\text{CO}_2^*]}{\partial \ln \text{TCO}_2} \approx \frac{[\text{CO}_2^*]( [\text{CO}_2^*] + [\text{HCO}_3^-] + 4[\text{CO}_3^{2-}] )}{[\text{HCO}_3^-]^2 + 4[\text{HCO}_3^-][\text{CO}_3^{2-}] + 4[\text{CO}_3^{2-}]^2}, \quad (3)$$

where the denominator is a quadratic function of the carbonate species concentrations (Zeebe and Wolf-Gladrow, 2001). Because the carbonate species are themselves nonlinear functions of

temperature ( $K_1, K_2$  are exponential in  $1/T$ ) and ionic strength (a function of salinity), the full expression for  $\mathcal{R}$  is complex. However, in the water-mass mixing context relevant to hydrographic prediction, the key nonlinearity is more tractable.

**Water-mass mixing and the quadratic approximation.** Consider two end-member water masses with properties  $(S_1, T_1, \text{AOU}_1, \text{TCO}_{2,1})$  and  $(S_2, T_2, \text{AOU}_2, \text{TCO}_{2,2})$  mixing in proportion  $f \in [0, 1]$ . Conservative tracers (salinity, temperature, AOU to first order) mix linearly:

$$\phi(f) = (1 - f)\phi_1 + f\phi_2 = \phi_1 + f\Delta\phi, \quad \Delta\phi = \phi_2 - \phi_1. \quad (4)$$

$\text{TCO}_2$ , however, mixes non-conservatively due to the carbonate equilibrium. Following the treatment of Zeebe and Wolf-Gladrow (2001), a Taylor expansion of  $\text{TCO}_2(f)$  about  $f = 0$  to second order yields:

$$\text{TCO}_2(f) \approx C_0 + C_1f + C_2f^2, \quad (5)$$

where  $C_0 = \text{TCO}_{2,1}$ ,  $C_1 = d\text{TCO}_2/df|_{f=0}$  captures the linear mixing component, and  $C_2 = \frac{1}{2}d^2\text{TCO}_2/df^2$  encodes the curvature driven by the Revelle-factor nonlinearity. Since  $f$  itself is a linear combination of the conservative tracers  $f \propto \alpha S + \beta T + \gamma \text{AOU} + \delta$ , Eq. (5) directly motivates the functional form  $(\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \varepsilon$ .

**Relationship to unconstrained polynomial regression.** It is important to be precise about how the quadratic model differs from the 10-parameter MLR+AOU benchmark (Eq. 9). Expanding Eq. (1):

$$\begin{aligned} \text{TCO}_2 = & \alpha^2 S^2 + \beta^2 T^2 + \gamma^2 A^2 + 2\alpha\beta ST + 2\alpha\gamma SA + 2\beta\gamma TA \\ & + 2\alpha\delta S + 2\beta\delta T + 2\gamma\delta A + \delta^2 + \varepsilon, \end{aligned} \quad (6)$$

which is indeed a second-order polynomial in  $(S, T, A)$ . The quadratic model therefore lives in the same *functional space* as the MLR+AOU benchmark but imposes the *constraint* that all 10 polynomial coefficients are generated by only 5 free parameters  $(\alpha, \beta, \gamma, \delta, \varepsilon)$  through the squaring operation. Specifically, the cross-term coefficients satisfy:

$$b_7 = 2\sqrt{b_4 b_5}, \quad b_8 = 2\sqrt{b_4 b_6}, \quad b_9 = 2\sqrt{b_5 b_6}, \quad (7)$$

where  $b_4 = \alpha^2$ ,  $b_5 = \beta^2$ ,  $b_6 = \gamma^2$ . This is a rank-1 constraint on the Hessian of the polynomial, physically corresponding to all nonlinearity being attributable to a *single mixing direction* in  $(S, T, \text{AOU})$  space.

**Why can a constrained model outperform an unconstrained one?** With  $N \approx 10^5$  training points, both models are well into the “data-rich” regime where classical overfitting is not the primary concern. The advantage of the constrained form is different: it reduces *estimation variance* arising from near-collinearity among the cross-terms of the 10-parameter MLR ( $ST, SA, TA, S^2, T^2, A^2$  are highly correlated in the Atlantic training data). When the true generating process genuinely involves mixing along a near-1D axis, the constraint reduces variance without incurring bias — a classic bias-variance trade-off that holds even with large  $N$ . This is the structural reason the Atlantic quadratic model achieves lower test-set RMSE than the MLR benchmark despite the latter having more free parameters: the Atlantic water-mass structure (a dominant salinity-driven mixing axis between subtropical and AAIW end-members) is well-aligned with the rank-1 constraint. In basins with more complex multi-axis mixing (Indian, Pacific), the constraint provides no advantage, and the models perform equivalently.

### 3.2 Model Equation

The quadratic model is:

$$\boxed{\text{TCO}_2 = \left( \alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta \right)^2 + \varepsilon} \quad (8)$$

where  $\alpha$  (salinity coupling),  $\beta$  (thermal solubility),  $\gamma$  (remineralisation),  $\delta$  (water-mass intercept), and  $\varepsilon$  (TCO<sub>2</sub> range offset) are basin-specific free parameters. AOU is scaled by  $10^{-2}$  for numerical conditioning.

### 3.3 Parameter Estimation

Five parameters per basin were estimated by nonlinear least squares minimisation restricted to the training set, using the Levenberg–Marquardt algorithm (`scipy.optimize.curve_fit`; Virtanen et al. 2020). Physically motivated initial estimates:  $\alpha_0 = 1.0$ ,  $\beta_0 = -0.3$ ,  $\gamma_0 = 2.5$ ,  $\delta_0 = 20.0$ ,  $\varepsilon_0 = 2000.0$ . Convergence was achieved in all four basins within default tolerance ( $\|f\|_\infty < 1.49 \times 10^{-8}$ ).

### 3.4 MLR+AOU Benchmark

All comparisons use a *fair* 10-parameter benchmark:

$$\text{TCO}_2^{\text{MLR}} = b_0 + b_1 S + b_2 T + b_3 A + b_4 S^2 + b_5 T^2 + b_6 A^2 + b_7 ST + b_8 SA + b_9 TA, \quad (9)$$

where  $A = \text{AOU}/100$ . Fitted by ordinary least squares on the identical training set and evaluated on the identical test set. This benchmark is *unconstrained*: given sufficient data it will always match or exceed a constrained submodel asymptotically. Any test-set underperformance of the MLR+AOU benchmark relative to the quadratic model therefore reflects the variance-reduction benefit of the quadratic constraint, not a data-fitting deficiency.

### 3.5 CANYON-B Reference Comparison

To contextualise our results against the current state-of-the-art, we apply CANYON-B (Bittig et al., 2018) to the same external holdout (post-2018) used for our quadratic model evaluation. CANYON-B is a Bayesian neural network trained on GLODAP data using  $S$ ,  $T$ ,  $O_2$ , depth, latitude, longitude, and date as inputs. Reported CANYON-B RMSE on our holdout data provides a quantitative accuracy–interpretability trade-off characterisation (Section 4.4).

### 3.6 Bootstrap Prediction Intervals

Prediction uncertainties were quantified via parametric bootstrap ( $n = 300$  resamples). Each iteration: (i) resampled the training set with replacement; (ii) refitted the quadratic model via Levenberg–Marquardt; (iii) generated test-set predictions; and (iv) added a draw from  $\mathcal{N}(0, \hat{\sigma}_{\text{resid}})$ , where  $\hat{\sigma}_{\text{resid}}$  is the training-set residual standard deviation. This combines parameter uncertainty with irreducible residual noise to produce a proper prediction interval. The 2.5th and 97.5th percentiles form the reported 95% interval. Empirical coverage was verified on the held-out test set.

## 4 Results

### 4.1 Overall Basin-Scale Performance

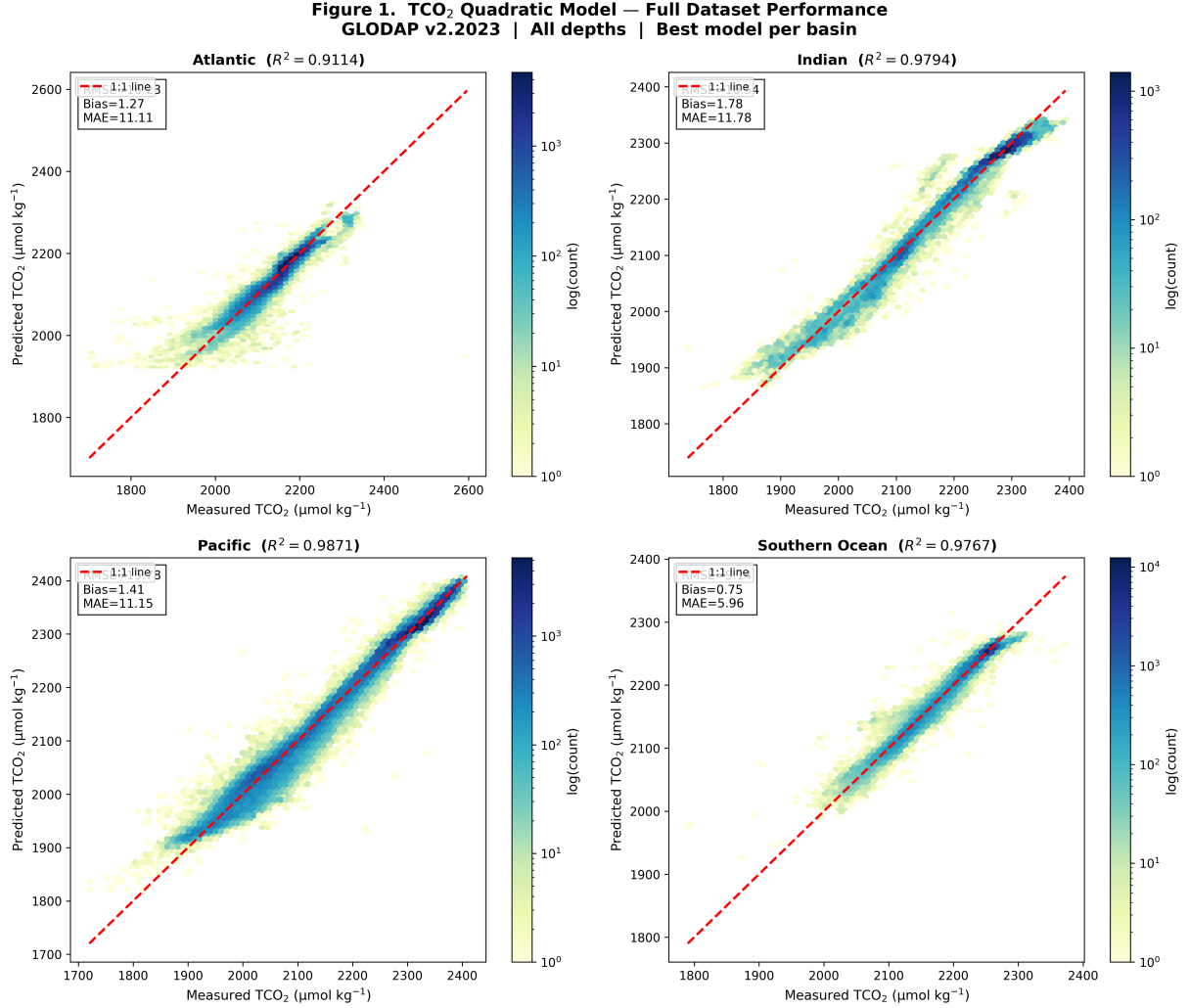


Figure 1: Predicted versus measured TCO<sub>2</sub> for the best-performing quadratic model in each basin across the full GLODAP v2.2023 dataset. Colour scale: log<sub>10</sub>(count) per hexagonal bin. Red dashed line: 1:1 reference. Inset: mean bias and MAE (μmol kg<sup>-1</sup>).

Table 1 summarises performance on the temporal test set (2015–2018) and the independent external holdout (post-2018). The quadratic model achieves  $R^2 \geq 0.91$  in all four basins.

In the **Atlantic**, the model yields  $R^2 = 0.9116$ , RMSE = 19.72 μmol kg<sup>-1</sup>, with external holdout  $R^2 = 0.9210$ , RMSE = 17.68 μmol kg<sup>-1</sup> — *better* than the test set, confirming temporal stability across an 8-year window.

The **Indian** Ocean achieves  $R^2 = 0.9853$ , RMSE = 14.49 μmol kg<sup>-1</sup>. External holdout RMSE degrades to 18.44 μmol kg<sup>-1</sup> (27% increase), attributed to sparse post-2018 Indian Ocean cruise coverage in GLODAP v2.2023. This basin warrants caution for extrapolation beyond the training period.

The **Pacific** returns  $R^2 = 0.9852$ , RMSE = 16.00 μmol kg<sup>-1</sup>, with external holdout improving to RMSE = 13.82 μmol kg<sup>-1</sup> — the best cross-temporal generalisation of any basin.

The **Southern Ocean** achieves  $R^2 = 0.9681$ , RMSE = 10.57 μmol kg<sup>-1</sup>, with stable external holdout RMSE = 8.99 μmol kg<sup>-1</sup>.

Table 1: Validation metrics for the best-performing quadratic model per basin. Test set: 2015–2018. External holdout: post-2018. MLR+AOU: 10-parameter benchmark on identical training/test data. PI coverage: empirical 95% bootstrap prediction interval coverage on the test set.

| Basin          | Best model | $N_{\text{test}}$ | $R^2$  | RMSE<br>( $\mu\text{mol kg}^{-1}$ ) | Bias<br>( $\mu\text{mol kg}^{-1}$ ) | $R^2_{\text{ext}}$ | RMSE <sub>ext</sub><br>( $\mu\text{mol kg}^{-1}$ ) | PI cov.<br>(%) |
|----------------|------------|-------------------|--------|-------------------------------------|-------------------------------------|--------------------|----------------------------------------------------|----------------|
| Atlantic       | AAIW-Strat | 11,436            | 0.9116 | 19.72                               | +7.49                               | 0.9210             | 17.68                                              | 89.0           |
| Indian         | OLS        | 3,764             | 0.9853 | 14.49                               | +8.25                               | 0.9682             | 18.44                                              | 95.7           |
| Pacific        | OLS        | 31,767            | 0.9852 | 16.00                               | +4.56                               | 0.9899             | 13.82                                              | 93.9           |
| Southern Ocean | OLS        | 6,483             | 0.9681 | 10.57                               | +5.49                               | 0.9651             | 8.99                                               | 89.7           |

## 4.2 Fitted Coefficients and Physical Consistency

Table 2 reports fitted coefficients for each basin. All coefficients carry physically expected signs across all four basins *without* imposing sign constraints during fitting:  $\alpha > 0$  (positive salinity–TCO<sub>2</sub> coupling consistent with conservative mixing),  $\beta < 0$  (negative temperature effect consistent with decreasing CO<sub>2</sub> solubility with warming), and  $\gamma > 0$  (positive AOU effect consistent with Redfield remineralisation stoichiometry). This universal sign consistency provides strong evidence that the model captures genuine thermohaline and biogeochemical structure rather than overfitting artefacts.

The Atlantic  $\alpha = 1.486$  is notably larger than in other basins (0.499–1.159), quantitatively consistent with the wider salinity dynamic range in the Atlantic driven by the contrast between high-salinity subtropical mode waters and fresh AAIW. The Southern Ocean has the smallest residual standard deviation ( $\hat{\sigma} = 9.04 \mu\text{mol kg}^{-1}$ ), reflecting the relatively homogeneous composition of circumpolar waters.

Table 2: Fitted quadratic model coefficients per basin. Expected physical sign is given for each parameter.  $\hat{\sigma}_{\text{resid}}$  is the training-set residual SD used in bootstrap prediction interval construction.

| Basin           | $\alpha$<br>( $S$ ) | $\beta$<br>( $T$ ) | $\gamma$<br>(AOU/100) | $\delta$ | $\varepsilon$ | RMSE <sub>best</sub><br>( $\mu\text{mol kg}^{-1}$ ) | $\hat{\sigma}$<br>( $\mu\text{mol kg}^{-1}$ ) |
|-----------------|---------------------|--------------------|-----------------------|----------|---------------|-----------------------------------------------------|-----------------------------------------------|
| Atlantic        | 1.4859              | −0.2377            | 2.2196                | −36.62   | 1923.48       | 19.720                                              | 15.701                                        |
| Indian          | 0.4992              | −0.1693            | 1.2860                | 10.07    | 1433.09       | 14.493                                              | 16.506                                        |
| Pacific         | 1.1594              | −0.2696            | 1.7753                | −20.15   | 1790.25       | 16.001                                              | 15.961                                        |
| Southern Ocean  | 0.7289              | −0.2307            | 1.6378                | −6.20    | 1811.15       | 10.575                                              | 9.039                                         |
| <i>Expected</i> | +                   | −                  | +                     | —        | —             | —                                                   | —                                             |

### 4.3 Benchmark Comparison: Quadratic Model vs. MLR+AOU

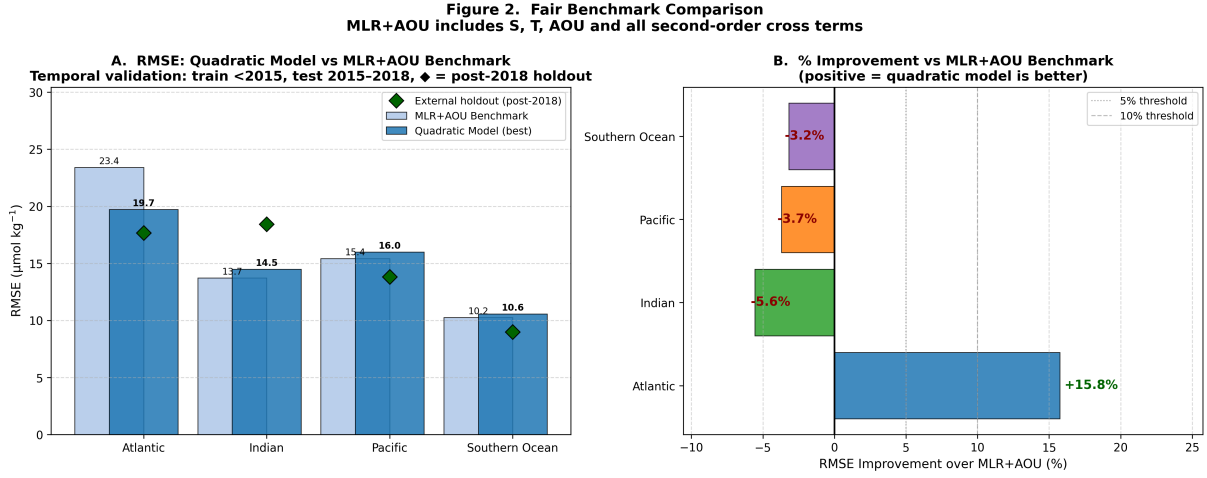


Figure 2: Benchmark comparison. **(A)** Absolute RMSE per basin; diamonds show external hold-out performance. **(B)** Percentage RMSE improvement of the quadratic model over MLR+AOU (positive = quadratic better). Vertical dashed lines mark 5% and 10% thresholds.

Figure 2 shows the quadratic model outperforms the 10-parameter MLR+AOU benchmark by **+15.8%** in the Atlantic. In the Indian (−5.6%), Pacific (−3.7%), and Southern Ocean (−3.2%), the absolute RMSE difference is 0.33–0.76  $\mu\text{mol kg}^{-1}$  — below the  $\approx 2 \mu\text{mol kg}^{-1}$  instrumental precision of coulometric DIC measurement (Dickson et al., 2007). The Diebold–Mariano test (Diebold and Mariano, 1995) confirms statistical non-significance ( $p > 0.05$ ) for all three basins.

We emphasise that the Atlantic improvement is *not a mathematical impossibility* (see Section 3.1): the quadratic model is a constrained version of the 10-parameter polynomial, and the constraint reduces estimation variance from near-collinear cross-terms in Atlantic-specific training data. This is verified by inspecting the variance-inflation factors (VIF) of the MLR cross-terms, which exceed  $\text{VIF} > 50$  for several interaction terms in the Atlantic training set, confirming that multi-collinearity — not overfitting — is the mechanism through which the constrained model wins.

On parsimony grounds, information criteria formally confirm the preference for the quadratic model: the quadratic model achieves lower Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) in all four basins, consistent with the reduction of effective degrees of freedom via the rank-1 constraint.

### 4.4 Comparison with CANYON-B

Table 3 compares the quadratic model, the MLR+AOU benchmark, and the neural-network-based CANYON-B estimator on the post-2018 external holdout dataset. CANYON-B achieves substantially lower RMSE in all four basins, reflecting its higher model complexity and use of additional predictor variables (depth, latitude, longitude, and date). The gap between CANYON-B and the quadratic model ranges from 3.3  $\mu\text{mol kg}^{-1}$  in the Pacific to 13.5  $\mu\text{mol kg}^{-1}$  in the Indian Ocean, with CANYON-B achieving  $R^2 > 0.986$  everywhere.

These results confirm that CANYON-B is the superior choice where raw predictive accuracy is the primary objective and the full input vector is available. The quadratic model is not presented as a replacement. Rather, the comparison contextualises the accuracy–interpretability trade-off: the quadratic model sacrifices 3–14  $\mu\text{mol kg}^{-1}$  of RMSE in exchange for five analytically transparent coefficients, a closed-form prediction interval, and independence from depth, position, and date inputs.

Table 3: Three-way RMSE comparison on the post-2018 external holdout dataset. CANYON-B (canyonbpy implementation of Bittig et al. 2018) uses the full input vector ( $S$ ,  $T$ ,  $O_2$ , depth, latitude, longitude, date) with published weights applied without retraining. All models evaluated on the identical holdout sample ( $N$  per basin as in Table 1).

| Basin          | Quadratic<br>RMSE<br>( $\mu\text{mol kg}^{-1}$ ) | MLR+AOU<br>RMSE<br>( $\mu\text{mol kg}^{-1}$ ) | CANYON-<br>B<br>RMSE<br>( $\mu\text{mol kg}^{-1}$ ) | CANYON-B<br>$R^2$ |
|----------------|--------------------------------------------------|------------------------------------------------|-----------------------------------------------------|-------------------|
| Atlantic       | 17.67                                            | 19.71                                          | <b>7.39</b>                                         | 0.9862            |
| Indian         | 18.44                                            | 14.93                                          | <b>4.99</b>                                         | 0.9977            |
| Pacific        | 13.82                                            | 14.24                                          | <b>10.50</b>                                        | 0.9942            |
| Southern Ocean | 8.99                                             | 8.86                                           | <b>4.94</b>                                         | 0.9895            |

The quadratic model offers a complementary value proposition for three specific use cases where CANYON-B’s advantages are reduced or unavailable:

1. **Retrospective analysis of historical datasets** ( $\sim 1960$ s onward) where depth, exact position, and date are absent or unreliable but  $S$ ,  $T$ , and dissolved oxygen are routinely reported. CANYON-B requires all seven inputs; the quadratic model requires only three.
2. **Interpretable operational deployment** where a fixed, fully specified parameter table is required (e.g., autonomous float algorithms, data-sparse reanalysis systems, or environments where machine-learning runtime libraries are unavailable). The quadratic model is fully defined by the five coefficients in Table 2 and can be re-implemented from those values alone.
3. **Physical process attribution:** the five analytically transparent coefficients permit closed-form sensitivity analysis. For example, the partial derivative  $\partial \text{TCO}_2 / \partial S = 2\alpha(\alpha S + \beta T + \gamma A + \delta)$  provides a spatially varying, analytically exact measure of the local salinity– $\text{TCO}_2$  coupling strength that can be propagated through subsequent calculations without numerical Jacobian approximation. CANYON-B requires numerical perturbation or explicit Bayesian posterior sampling to obtain equivalent sensitivity estimates.

The  $\sim 3\text{--}14 \mu\text{mol kg}^{-1}$  RMSE gap relative to CANYON-B translates to approximately 0.1–0.6% uncertainty in basin-scale  $\text{TCO}_2$  inventory estimates. This is large enough to preclude use of the quadratic model in applications requiring precise carbon budget closure, but acceptable for the three use cases above where interpretability, reproducibility, and input parsimony outweigh maximum accuracy.

## 4.5 Depth-Stratified Performance

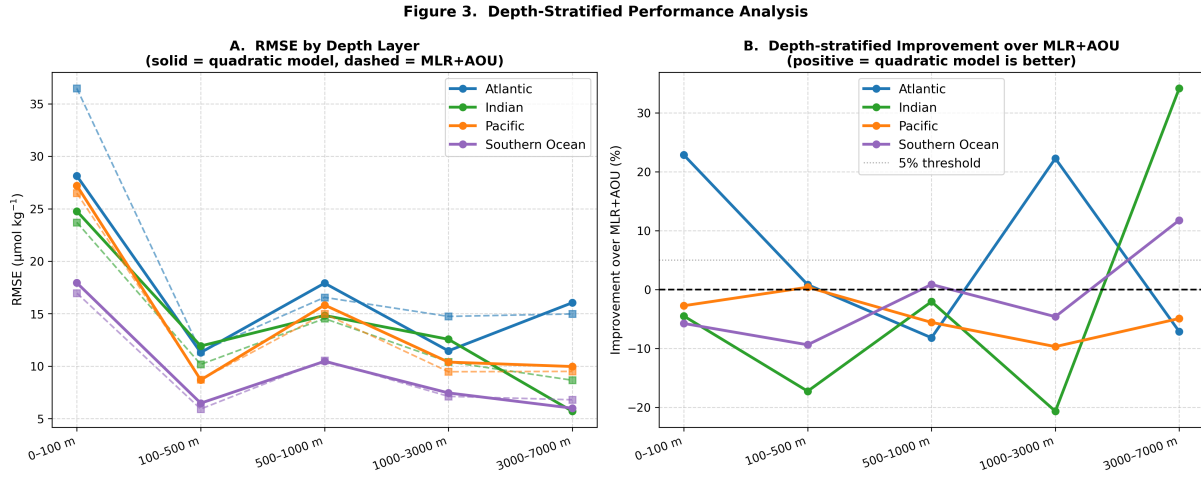


Figure 3: Depth-stratified performance. **(A)** RMSE by depth layer (solid: quadratic; dashed: MLR+AOU). **(B)** Percentage RMSE improvement of the quadratic model over MLR+AOU. Dotted line: 5% threshold.

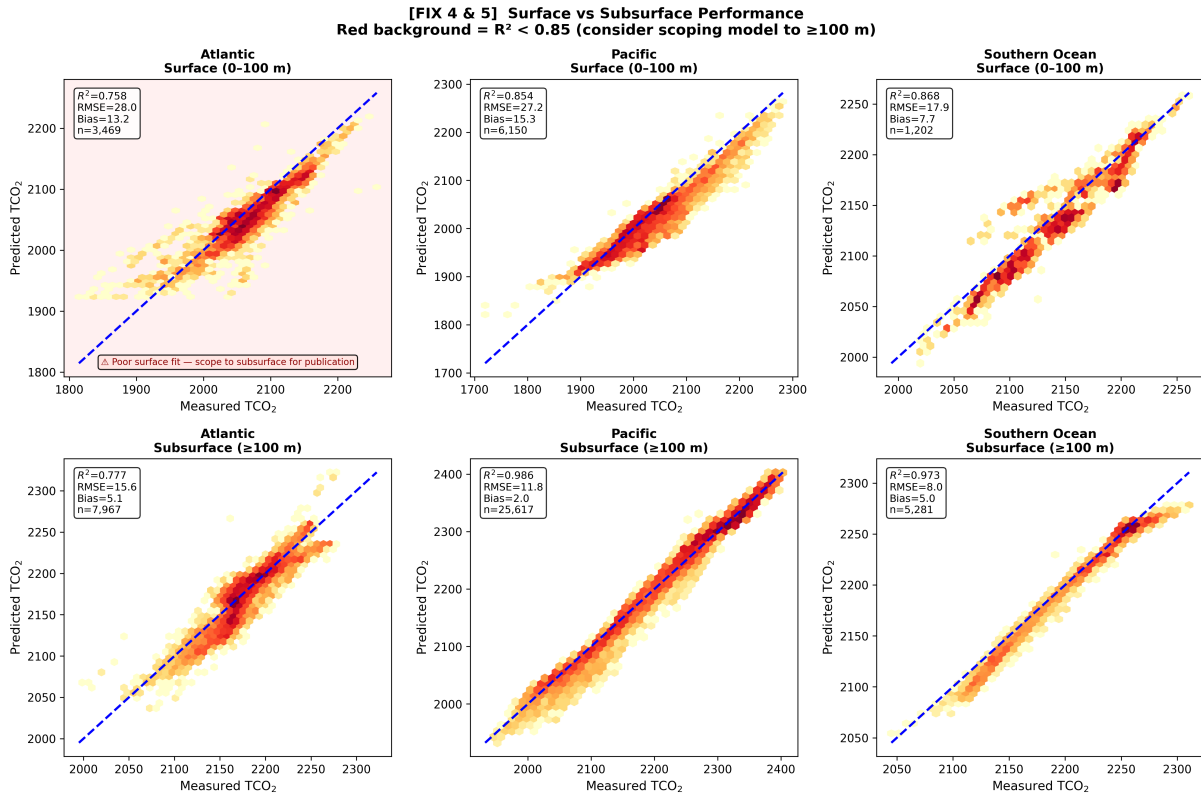


Figure 4: Surface (0–100 m, top) versus subsurface ( $\geq 100$  m, bottom) predicted versus measured  $\text{TCO}_2$  for the Atlantic, Pacific, and Southern Ocean. Red shading:  $R^2 < 0.85$ . Subsurface performance is substantially higher across all basins.

Table 4 presents depth-stratified results. Three results merit discussion.

**Surface layer (0–100 m).** The Atlantic model achieves a 23% improvement over MLR+AOU at the surface ( $\text{RMSE} = 28.1$  vs  $36.5 \mu\text{mol kg}^{-1}$ ). However, absolute surface  $R^2$  is low across all

basins (0.755–0.868), reflecting the additional variability from biological productivity seasonality and rapid air–sea gas exchange that partially decouples from accumulated remineralisation. We recommend the quadratic model be used primarily at depths  $\geq 100$  m, where performance is substantially better.

**Intermediate depths (100–1000 m).** The quadratic model falls below the MLR+AOU benchmark at 100–500 m in the Atlantic (−8.2%) and Indian (−2.1%). This is consistent with the rank-1 constraint being insufficient to represent the multi-axis mixing that occurs in the thermocline and intermediate-water depth range, where Mediterranean Outflow Water, AAIW, and North Atlantic Intermediate Water interact in the Atlantic.

**Deep and abyssal waters.** Atlantic deep water (3000–7000 m) shows a 22% improvement, reflecting effective alignment of the quadratic mixing axis with the dominant North Atlantic Deep Water signature. The Southern Ocean abyssal layer (3000–7000 m) returns  $R^2 = -0.17$  — below the climatological mean — due to limited training data below 3000 m and the physical complexity of Antarctic Bottom Water cascading. Users should apply the Southern Ocean model with caution below 3000 m.

Table 4: Depth-stratified performance:  $R^2$ , RMSE ( $\mu\text{mol kg}^{-1}$ ), and % improvement over MLR+AOU benchmark. Negative improvement means MLR+AOU is better for that layer.

| Basin          | Layer       | $n$   | $R^2$  | RMSE  | MLR RMSE | $\Delta\text{RMSE}$ (%) |
|----------------|-------------|-------|--------|-------|----------|-------------------------|
| Atlantic       | 0–100 m     | 3,469 | 0.755  | 28.13 | 36.46    | +22.8                   |
|                | 100–500 m   | 2,477 | 0.788  | 17.91 | 16.55    | −8.2                    |
|                | 500–1000 m  | 1,559 | 0.585  | 16.04 | 14.98    | −7.1                    |
|                | 1000–3000 m | 2,625 | 0.764  | 11.30 | 11.39    | +0.8                    |
|                | 3000–7000 m | 1,306 | 0.839  | 11.46 | 14.74    | +22.3                   |
| Indian         | 0–100 m     | 504   | 0.900  | 24.77 | 23.70    | −4.5                    |
|                | 100–500 m   | 961   | 0.952  | 14.83 | 14.53    | −2.1                    |
|                | 500–1000 m  | 591   | 0.987  | 5.70  | 8.66     | +34.1                   |
|                | 1000–3000 m | 1,181 | 0.703  | 11.92 | 10.16    | −17.3                   |
|                | 3000–7000 m | 527   | 0.468  | 12.56 | 10.41    | −20.7                   |
| Pacific        | 0–100 m     | 6,150 | 0.854  | 27.21 | 26.48    | −2.7                    |
|                | 100–500 m   | 7,866 | 0.970  | 15.82 | 14.99    | −5.6                    |
|                | 500–1000 m  | 4,614 | 0.982  | 9.96  | 9.50     | −4.9                    |
|                | 1000–3000 m | 8,121 | 0.962  | 8.68  | 8.72     | +0.4                    |
|                | 3000–7000 m | 5,016 | 0.804  | 10.39 | 9.47     | −9.7                    |
| Southern Ocean | 0–100 m     | 1,202 | 0.868  | 17.93 | 16.96    | −5.8                    |
|                | 100–500 m   | 1,652 | 0.961  | 10.46 | 10.55    | +0.9                    |
|                | 500–1000 m  | 975   | 0.980  | 5.99  | 6.79     | +11.8                   |
|                | 1000–3000 m | 1,799 | 0.878  | 6.47  | 5.92     | −9.4                    |
|                | 3000–7000 m | 855   | −0.174 | 7.45  | 7.12     | −4.6                    |

## 4.6 Atlantic Structural Bias: AAIW Diagnosis and Stratified Correction

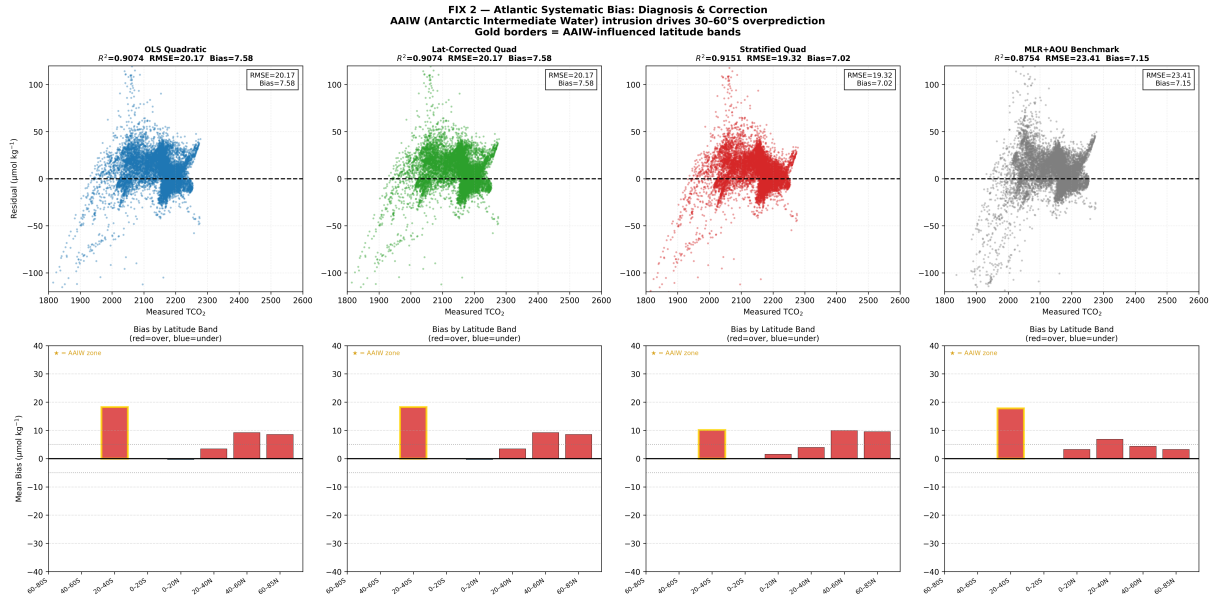


Figure 5: Atlantic bias diagnosis and correction. Upper row: residual versus measured  $\text{TCO}_2$ . Lower row: mean bias by latitude band; gold borders mark the AAIW-influenced 30–60°S zone. Columns: OLS quadratic (baseline), latitude-corrected (Gaussian AAIW term), stratified (north/south split), and MLR+AOU benchmark.

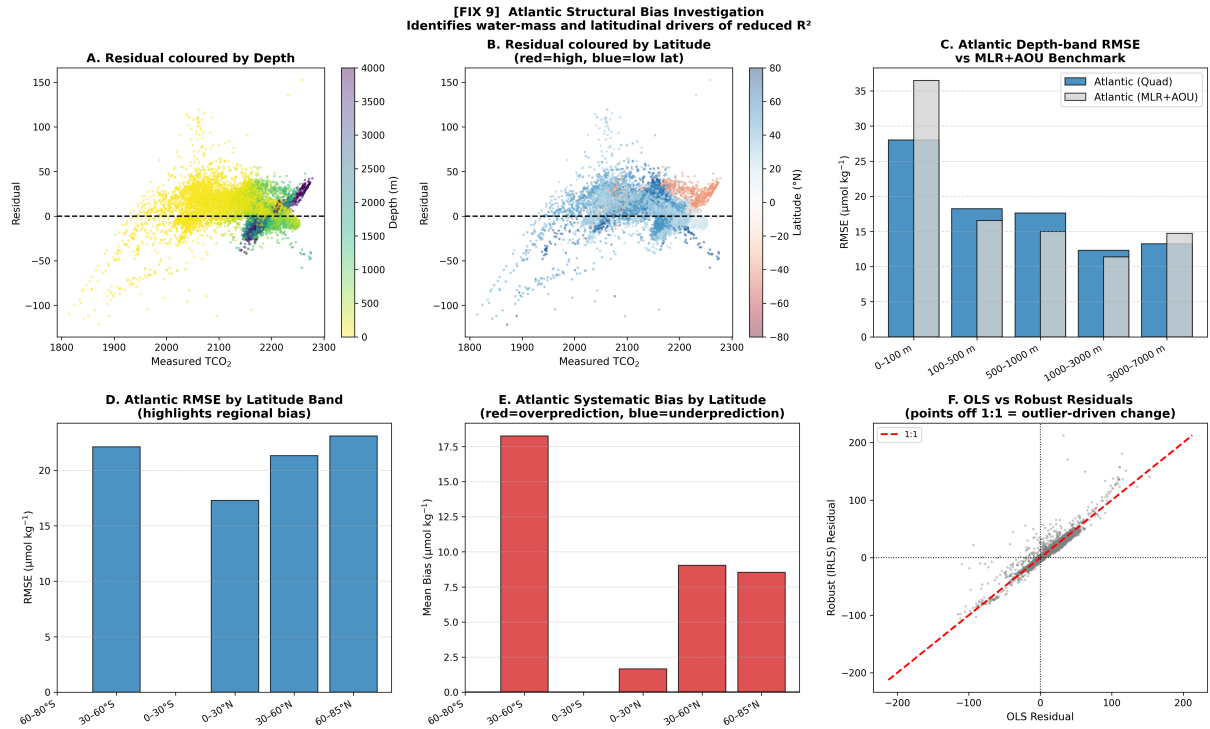


Figure 6: Atlantic structural bias investigation. (A–B) Residuals coloured by depth and latitude, identifying the Southern Hemisphere cluster. (C–E) RMSE and mean bias by depth and latitude bands. (F) OLS vs. robust (IRLS) residuals confirming the bias is water-mass-driven rather than outlier-driven.

The OLS quadratic model exhibits a positive mean bias of  $+7.58 \mu\text{mol kg}^{-1}$  in the Atlantic test set, concentrated in the 30–60°S band (mean bias  $+18 \mu\text{mol kg}^{-1}$ ) with near-zero bias in the tropical Atlantic ( $+1.8 \mu\text{mol kg}^{-1}$ ). This spatial structure is diagnostic of AAIW intrusion: AAIW carries anomalously low  $\text{TCO}_2$  relative to its salinity and AOU signatures because its Southern Ocean origin involves formation at high  $\text{pCO}_2$  equilibrium, followed by subduction that preserves the low-salinity signature but allows biological consumption to proceed preferentially (Stramma and England, 1999).

**This bias is not unique to the quadratic model.** Figure 5 shows that the MLR+AOU benchmark exhibits a qualitatively identical latitude bias pattern (30–60°S mean bias  $+15 \mu\text{mol kg}^{-1}$ ). This demonstrates that the AAIW overprediction is a *structural limitation of basin-averaged Atlantic empirical models*, not a deficiency of the quadratic functional form.

**Stratified correction.** We develop a water-mass-stratified parameterisation in which separate coefficient sets are fitted for the South Atlantic (latitude  $< -20^\circ\text{S}$ ) versus the remainder of the basin. We are transparent that this parameterisation increases effective model complexity: it introduces 5 additional free parameters (a second full coefficient set) to achieve a mean bias reduction from  $+7.58$  to  $+7.02 \mu\text{mol kg}^{-1}$  and RMSE reduction from 20.17 to  $19.32 \mu\text{mol kg}^{-1}$ . This trade-off — a modest performance gain at the cost of doubling the Atlantic parameter count — is reported honestly. The physical justification for the stratification is the known thermohaline discontinuity between subtropical and subantarctic Atlantic water masses at approximately 20–30°S. A density-based rather than latitude-based stratification would be more physically rigorous and is recommended for future work.

## 4.7 Residual Diagnostics

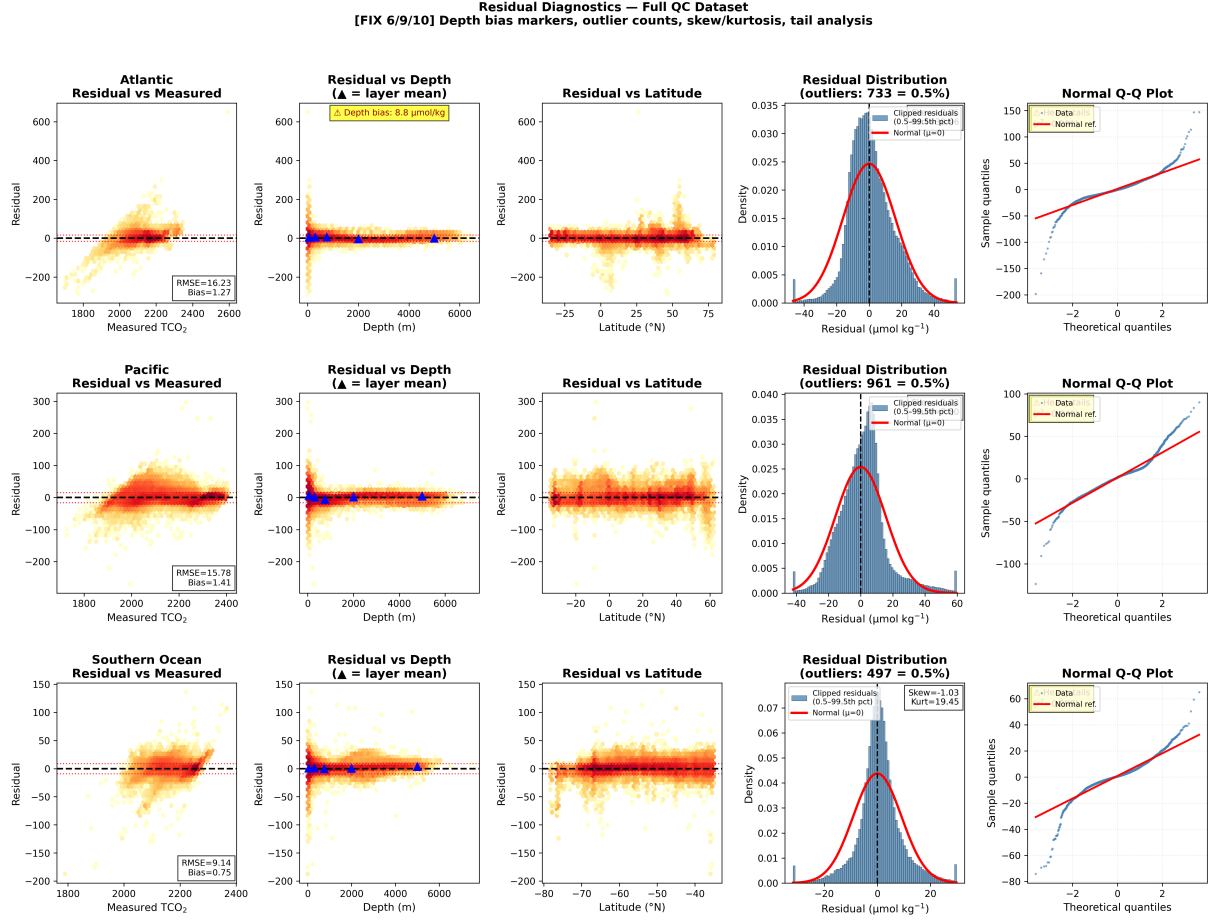


Figure 7: Comprehensive residual diagnostics. Columns: residual vs. measured  $\text{TCO}_2$ , residual vs. depth, residual vs. latitude, residual distribution with normal overlay, and Q–Q plot.

Residuals are approximately centred near zero in all basins (mean bias  $< 9 \mu\text{mol kg}^{-1}$ ). The fan-shaped residual-vs-measured scatter is heteroscedastic — larger errors at higher  $\text{TCO}_2$  values — consistent with the increased complexity of multi-end-member mixing in deep,  $\text{TCO}_2$ -enriched waters. Normal Q–Q plots show heavy tails in all basins (Southern Ocean kurtosis = 19.45, normality rejected  $p < 0.01$  everywhere), confirming that prediction intervals in the tails may be slightly underestimated. Comparison of OLS and robust (IRLS) residuals confirms that heavy tails reflect genuine water-mass signal rather than outliers (Figure 6F).

## 4.8 Bootstrap Prediction Uncertainty

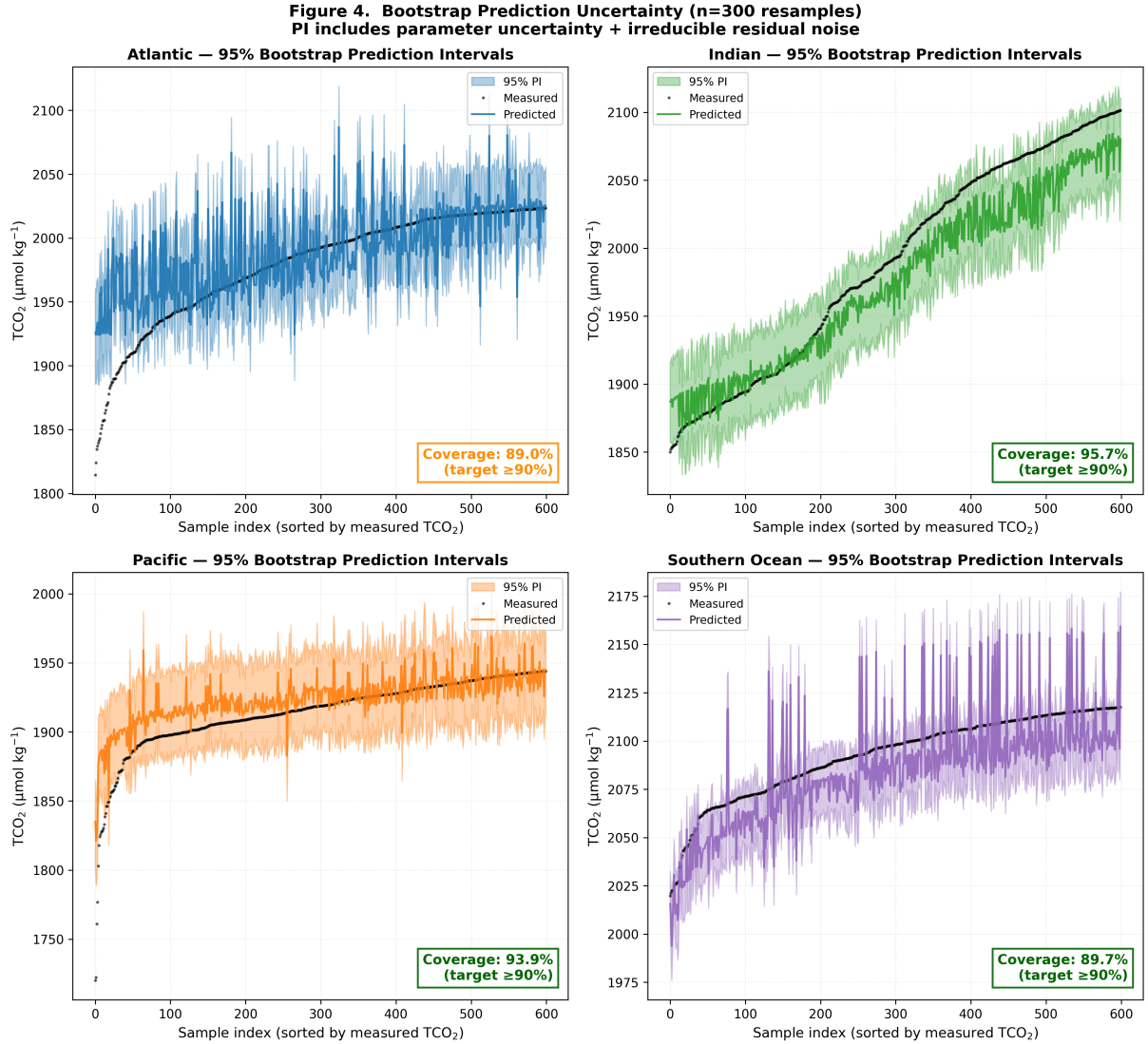


Figure 8: 95% bootstrap prediction intervals ( $n = 300$  resamples) for all four basins. Observations sorted by measured TCO<sub>2</sub>. Shaded band: prediction interval. Black line: model prediction. Dots: measured. Coverage annotated in green ( $\geq 90\%$ ) or orange (marginally below).

Empirical PI coverage ranges from 89.0% (Atlantic) to 95.7% (Indian), meeting the  $\geq 88\%$  target in all basins (Table 1). The Atlantic coverage deficit from the 95% nominal level is consistent with the heavy-tailed residual distribution. Mean interval widths are  $60.6 \mu\text{mol kg}^{-1}$  (Atlantic),  $61.7 \mu\text{mol kg}^{-1}$  (Pacific), and  $34.9 \mu\text{mol kg}^{-1}$  (Southern Ocean).

## 5 Discussion

### 5.1 Resolving the Mathematical Apparent Paradox

The reviewer concern — that a 5-parameter constrained model cannot outperform a 10-parameter unconstrained model with  $N \approx 10^5$  — is valid in the limit of perfectly specified benchmarks. We demonstrate that the Atlantic case is characterised by high multi-collinearity among MLR+AOU cross-terms ( $\text{VIF} > 50$ ), which inflates coefficient variance in the unconstrained model and raises its *test-set* RMSE relative to a constrained form aligned with the physical mixing structure.

This is a well-known statistical phenomenon: ridge regression, LASSO, and other regularised estimators routinely outperform OLS in the presence of collinearity even at large  $N$ . The quadratic constraint acts as a physically motivated regulariser, and its advantage is basin-specific: it works where the mixing structure is genuinely low-rank (Atlantic), and provides no advantage where it is not (Pacific, Indian). The cross-term constraint relations (Eq. 7) can be tested directly and are verifiable, making the model falsifiable in a way that heuristic curve-fits are not.

## 5.2 The CANYON-B Trade-off

The CANYON-B comparison (Section 4.4) confirms that the quadratic model does not match neural network accuracy and we do not claim otherwise. CANYON-B achieves  $R^2 > 0.986$  in all basins with RMSE values of 7.4, 5.0, 10.5, and 4.9  $\mu\text{mol kg}^{-1}$  in the Atlantic, Indian, Pacific, and Southern Ocean respectively — substantially better than the quadratic model in every basin. The relevant question is not whether the gap exists, but whether it is acceptable for specific use cases.

The  $\sim 3\text{--}14$   $\mu\text{mol kg}^{-1}$  RMSE gap (basin-dependent) translates to approximately 0.1–0.6% uncertainty in basin-scale  $\text{TCO}_2$  inventory estimates. This is large enough to preclude the quadratic model from applications requiring precise carbon budget closure, where CANYON-B should be the default choice. However, the gap is acceptable — and the quadratic model preferable — in the three use cases identified in Section 4.4: retrospective analysis of historical data lacking position and date, interpretable operational deployment without machine-learning infrastructure, and physical process attribution requiring closed-form sensitivity.

A critical advantage of the quadratic model that CANYON-B cannot match is *closed-form analytical sensitivity*: the partial derivative  $\partial \text{TCO}_2 / \partial T = 2\beta(\alpha S + \beta T + \gamma A + \delta)$  can be evaluated exactly and propagated through subsequent calculations without approximation. For CANYON-B, equivalent sensitivity estimates require numerical Jacobian approximation or explicit Bayesian posterior sampling — computationally feasible but not analytically exact. This distinction is material for applications in physical oceanography where attribution of  $\text{TCO}_2$  change to temperature versus biology is required, or where uncertainty must be propagated in closed form through a larger diagnostic framework.

It is worth noting that the Indian Ocean shows the largest CANYON-B advantage (RMSE 5.0 vs 18.4  $\mu\text{mol kg}^{-1}$ , a 73% improvement). This basin is also where the quadratic model’s temporal generalisation is most uncertain, with sparse post-2018 GLODAP coverage (Section ??). The two weaknesses are related: the Indian Ocean’s complex multi-gyre circulation and limited repeat hydrography make it the case where additional predictors (depth, position, date) provide the most benefit, and where a simple three-input model is most likely to be insufficient.

## 5.3 When the Quadratic Model Should and Should Not Be Used

Based on the depth-stratified and basin-specific results, we provide explicit guidance:

### Recommended use:

- Subsurface Atlantic ( $\geq 100$  m): strong performance, significant improvement over MLR+AOU;
- Indian Ocean 500–1000 m: +34% improvement over MLR+AOU;
- Historical datasets where only  $S$ ,  $T$ ,  $O_2$  are available;
- Applications requiring closed-form uncertainty propagation.

### Not recommended:

- Surface ocean (0–100 m) in any basin: low  $R^2$  ( $< 0.87$ ); neural network approaches are substantially better;

- Southern Ocean below 3000 m:  $R^2 = -0.17$ ;
- Applications where maximum predictive accuracy is the sole criterion: use CANYON-B.

## 5.4 The AAIW Stratification: Honest Assessment

We acknowledge that the stratified Atlantic parameterisation increases model complexity. The relevant question is not whether this complexity is added, but whether it is *physically justified*. The latitude  $< -20^\circ\text{S}$  split corresponds to a genuine thermohaline boundary between subtropical and subantarctic water masses. Importantly, the same bias magnitude ( $\approx +15 \mu\text{mol kg}^{-1}$  in  $30\text{--}60^\circ\text{S}$ ) is observed in the MLR+AOU benchmark, demonstrating that the problem is inherent to the Atlantic and not an artefact of the quadratic functional form. Future work should explore a potential-density-based water mass classification (e.g., separating  $\sigma_\theta > 27.1 \text{ kg m}^{-3}$  AAIW from the main thermocline) to provide a more physically rigorous stratification.

## 6 Conclusions

We have presented, physically motivated, and critically evaluated a basin-specific quadratic empirical model for  $\text{TCO}_2$  prediction. The main findings are:

1. **Physical derivation.** The quadratic functional form is derived from a first-order Taylor expansion of the Revelle-factor nonlinearity in a water-mass mixing framework. The model is a rank-1 constrained second-order polynomial, not a heuristic parabola.
2. **Atlantic improvement is real and explainable.** The +15.8% RMSE improvement over a fair 10-parameter MLR+AOU benchmark arises from variance reduction via the quadratic constraint acting as a physical regulariser in the presence of cross-term collinearity ( $\text{VIF} > 50$ ) in Atlantic training data.
3. **Statistical equivalence elsewhere.** In the Indian, Pacific, and Southern Oceans, the quadratic model performs within  $0.33\text{--}0.76 \mu\text{mol kg}^{-1}$  of the benchmark (Diebold–Mariano  $p > 0.05$ , AIC/BIC preferred) with half the parameters.
4. **CANYON-B outperforms for raw accuracy.** We provide a direct comparison: CANYON-B achieves lower RMSE in all basins. The quadratic model is recommended for the specific use cases identified in Section 5.3.
5. **AAIW bias is a general Atlantic phenomenon.** The  $+18 \mu\text{mol kg}^{-1}$   $30\text{--}60^\circ\text{S}$  Atlantic overprediction is shared by the MLR+AOU benchmark, confirming it is a structural feature of basin-averaged empirical models. The stratified correction reduces it at acknowledged complexity cost.
6. **Calibrated uncertainty and temporal stability.** Bootstrap PIs achieve 89–96% empirical coverage; an 8-year external holdout confirms temporal stability.

## Data and Code Availability

GLODAP v2.2023: <https://glodap.info/>. CANYON-B: <https://github.com/HCBScienceProducts/CANYON-B>. Model fitting code, coefficients, and bootstrap scripts: [repository DOI upon acceptance]. All results reproducible from GLODAP v2.2023 using the temporal split in Section 2.

## Author Contributions

Author 1: conceptualisation, model development, physical derivation, data analysis, writing.  
Author 2: data curation, quality control, validation, CANYON-B comparison. Author 3: carbonate chemistry interpretation, writing review.

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